

Logistic regression

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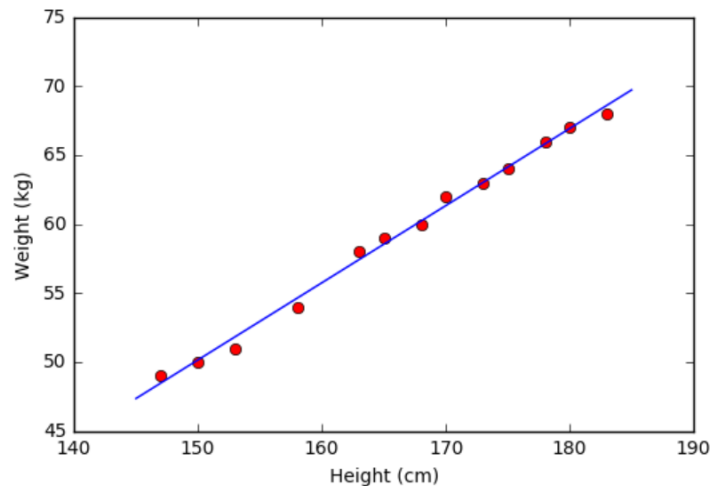
Introduction

Linear Regression and the Perceptron Learning Algorithm (PLA) share the same mathematical form: a linear combination of input features and weights

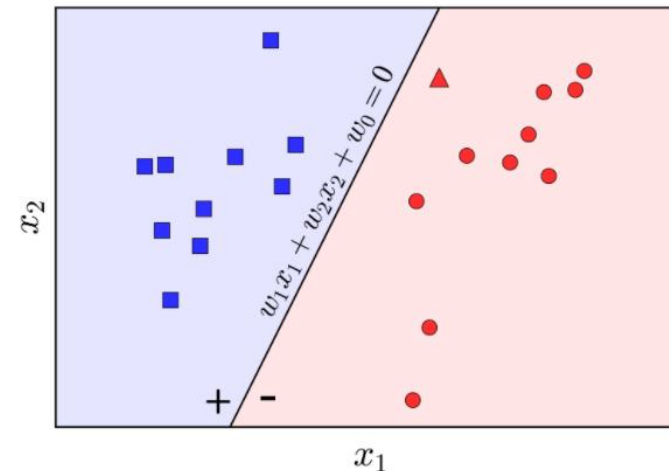
$$y = f(\mathbf{w}^T \mathbf{x})$$

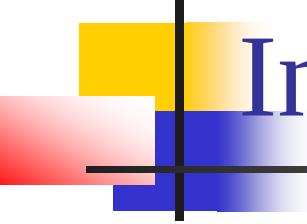
where $f(\mathbf{w}^T \mathbf{x})$ is an *activation function*

Linear regression: $f(\mathbf{w}^T \mathbf{x}) = \mathbf{w}^T \mathbf{x}$



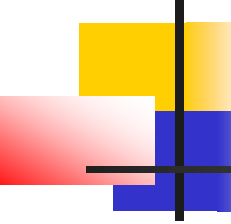
PLA $f: (\mathbf{w}^T \mathbf{x}) = \text{sgn}(\mathbf{w}^T \mathbf{x})$





Introduction

- Linear regression uses the dot product $w^T x$ to predict the output y . This method is suitable for problems where the output y is a real number without upper or lower bounds
- PLA, the output can take only one of two values, $+1$ or -1 . It is suitable for binary classification tasks

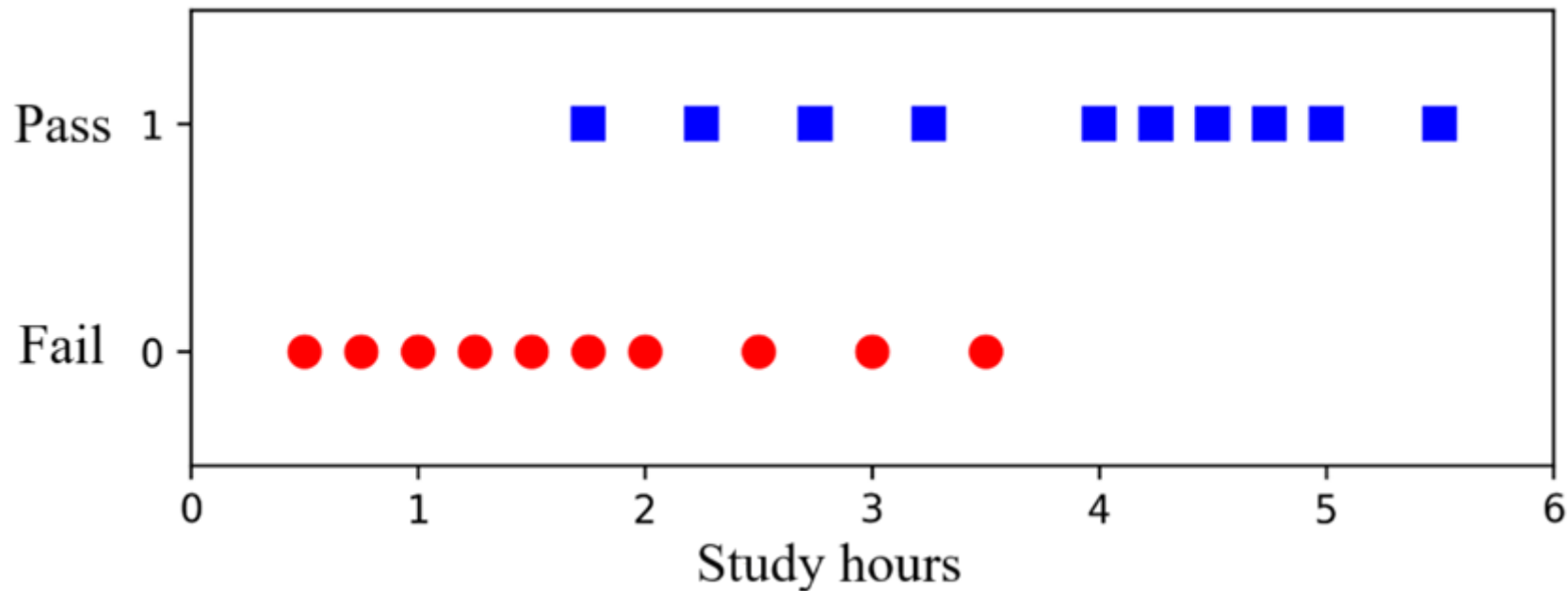


Introduction

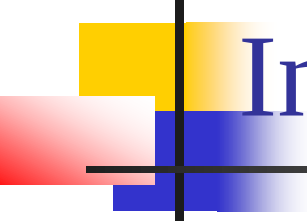
Given a group of 20 students spent between 0 and 6 hours preparing for an exam, with their results summarized in the following table. A student who studies for x hours will pass or fail?

Hours	Pass	Hours	Pass
.5	0	2.75	1
.75	0	3	0
1	0	3.25	1
1.25	0	3.5	0
1.5	0	4	1
1.75	0	4.25	1
1.75	1	4.5	1
2	0	4.75	1
2.25	1	5	1
2.5	0	5.5	1

Introduction



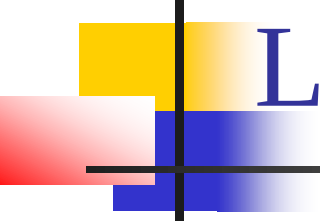
Neither linear regression nor PLA is appropriate for this problem; instead, a more flexible model is required



Introduction

This lecture introduces logistic regression:

- The output can be expressed in terms of probability. For example, the probability of passing conditioned on the number of study hours, or the probability of rain tomorrow based on today's observed weather information
- ***Logistic regression:***
 - is similar to linear regression in that the output is a real number
 - is similar to PLA in that the output is bounded (within the interval $[0,1]$)
- Although the term 'regression' appears in its name, logistic regression is more commonly used for classification tasks



Logistic Regression

The predicted output of:

- Linear Regression:

$$f(\mathbf{x}) = \mathbf{w}^T \mathbf{x}$$

- PLA:

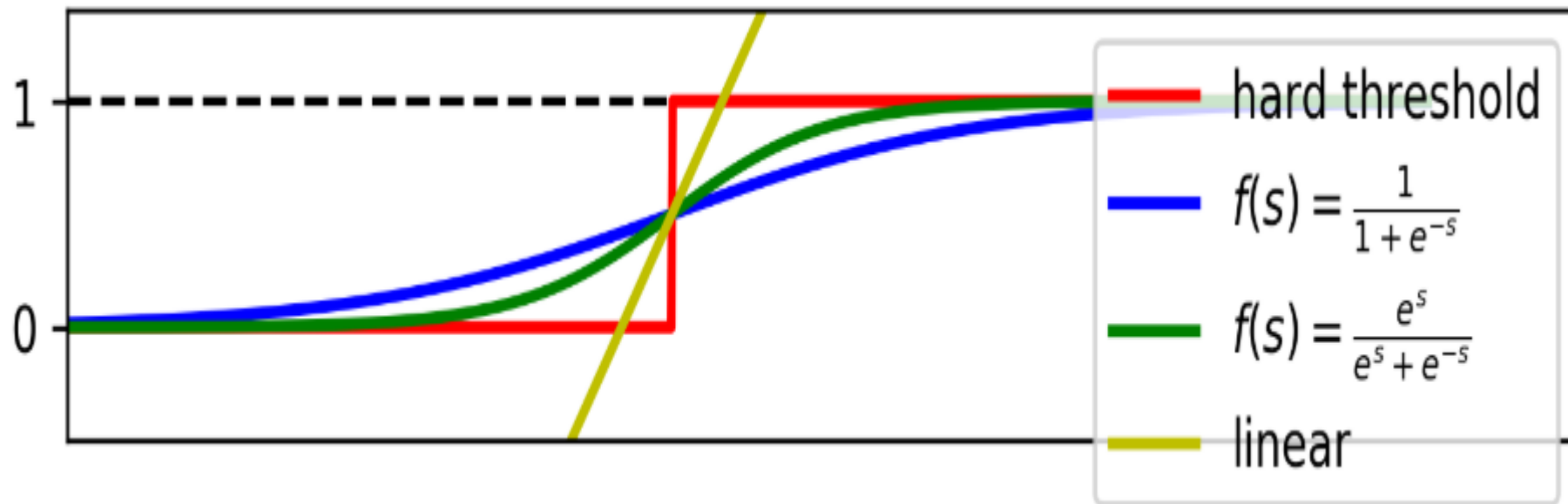
$$f(\mathbf{x}) = \text{sgn}(\mathbf{w}^T \mathbf{x})$$

- Logistic regression

$$f(\mathbf{x}) = \theta(\mathbf{w}^T \mathbf{x})$$

Logistic Regression

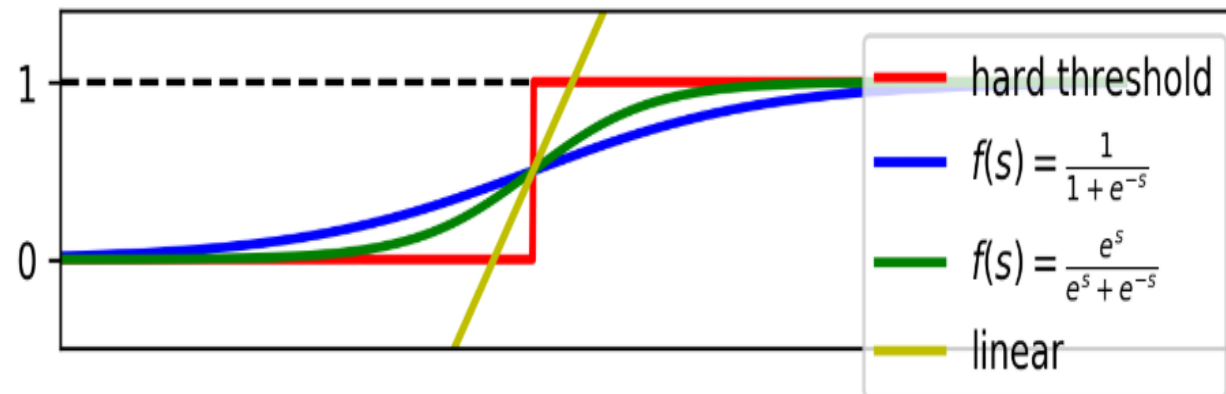
- Some activation functions for the linear model:

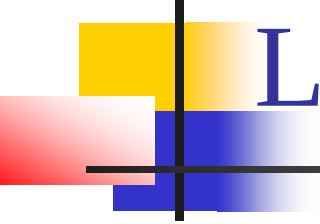


Logistic Regression

The blue and green curves better fit our problem for the following reasons:

- They are continuous real-valued functions, bounded within the interval (0,1)
- Taking 1/2 as the decision threshold: points farther to the left approach 0, while points farther to the right approach 1. This aligns with the intuition that more study hours lead to a higher probability of passing
- They are smooth and differentiable everywhere, which can be advantageous for optimization





Logistic Regression

- Sigmoid function

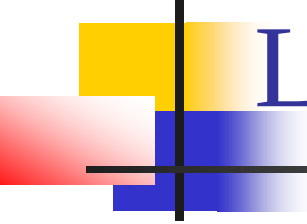
$$f(s) = \frac{1}{1 + e^{-s}} \triangleq \sigma(s)$$

- This function is most commonly used for the following reasons:
 - Bounded within the interval (0,1)
 - Has finite limits:

$$\lim_{s \rightarrow -\infty} \sigma(s) = 0; \quad \lim_{s \rightarrow +\infty} \sigma(s) = 1$$

- Simple derivative

$$\sigma'(s) = \sigma(s)(1 - \sigma(s))$$

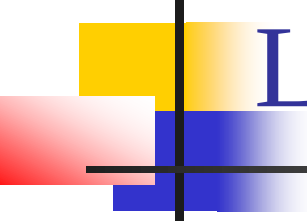


Loss function

- Assume that the probability of a data point $\mathbf{x}$ belonging to class 1 is $f(\mathbf{w}^T \mathbf{x})$ and belonging to class 0 is $(1 - f(\mathbf{w}^T \mathbf{x}))$
- For a training data point (with known output y), we can write:

$$\begin{aligned}P(y_i = 1 | \mathbf{x}_i; \mathbf{w}) &= f(\mathbf{w}^T \mathbf{x}_i) \\P(y_i = 0 | \mathbf{x}_i; \mathbf{w}) &= 1 - f(\mathbf{w}^T \mathbf{x}_i)\end{aligned}$$

- Find $\mathbf{w}$ so that $f(\mathbf{w}^T \mathbf{x}_i) \approx 1$ for class 1 points, and $f(\mathbf{w}^T \mathbf{x}_i) \approx 0$ for class 0 points



Loss function

- Set $z_i = f(\mathbf{w}^T \mathbf{x}_i)$ and combine the two expressions:

$$P(y_i = 1 | \mathbf{x}_i; \mathbf{w}) = f(\mathbf{w}^T \mathbf{x}_i)$$

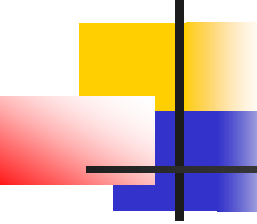
$$P(y_i = 0 | \mathbf{x}_i; \mathbf{w}) = 1 - f(\mathbf{w}^T \mathbf{x}_i)$$

we obtain:

$$P(y_i | \mathbf{x}_i; \mathbf{w}) = z_i^{y_i} (1 - z_i)^{1-y_i}$$

- This expression is equivalent to the previous two:
 - if $y_i = 1$ the second term on the right-hand side vanishes
 - if $y_i = 0$, the first term vanishes
- We want the model to fit the given data as closely as possible, i.e., to maximize the probability

$$P(y_i | \mathbf{x}_i; \mathbf{w})$$

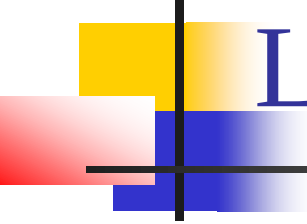


Loss function

- Considering the entire training set with $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N] \in \mathbf{R}^{d \times N}$ and $y = [y_1, y_2, \dots, y_N]$, we need to find $\mathbf{w}$ that maximizes $P(\mathbf{y}|\mathbf{X}; \mathbf{w})$

$$\mathbf{w} = \arg \max_{\mathbf{w}} P(\mathbf{y}|\mathbf{X}; \mathbf{w})$$

- The problem of finding the parameters so that the model fits the data as closely as possible is generally known as the maximum likelihood estimation (MLE) problem, where the function following the arg max is called the likelihood function

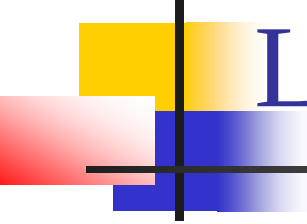


Loss function

- $z_i = f(\mathbf{w}^T \mathbf{x}_i) = \text{sigmoid}(\mathbf{w}^T \mathbf{x}_i)$, $P(y_i | \mathbf{x}_i; \mathbf{w}) = z_i^{y_i} (1 - z_i)^{1-y_i}$
- Assuming independent data points, we can write:

$$\begin{aligned} P(\mathbf{y} | \mathbf{X}; \mathbf{w}) &= \prod_{i=1}^N P(y_i | \mathbf{x}_i; \mathbf{w}) \\ &= \prod_{i=1}^N z_i^{y_i} (1 - z_i)^{1-y_i} \end{aligned}$$

- Directly optimizing this function w.r.t $\mathbf{w}$ is difficult

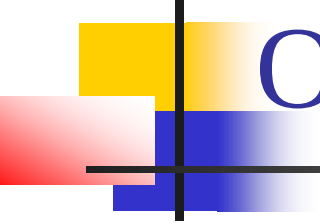


Loss function

$$\begin{aligned} P(\mathbf{y}|\mathbf{X}; \mathbf{w}) &= \prod_{i=1}^N P(y_i|\mathbf{x}_i; \mathbf{w}) \\ &= \prod_{i=1}^N z_i^{y_i} (1 - z_i)^{1-y_i} \end{aligned}$$

- Take the natural log of the likelihood function to convert products into sums and prevent very small values
- Taking the negative gives a loss function; maximizing likelihood becomes minimizing this negative log-likelihood

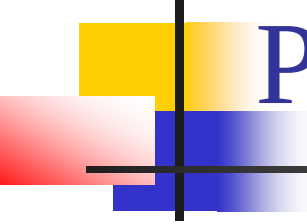
$$\begin{aligned} J(\mathbf{w}) &= -\log P(\mathbf{y}|\mathbf{X}; \mathbf{w}) \\ &= -\sum_{i=1}^N (y_i \log z_i + (1 - y_i) \log(1 - z_i)) \end{aligned}$$



Optimize the loss function

- Use gradient descent to optimize w
- Gradient descent update formula for logistic regression:

$$\mathbf{w} = \mathbf{w} + \eta(y_i - z_i)\mathbf{x}_i$$



Predict the class of point x

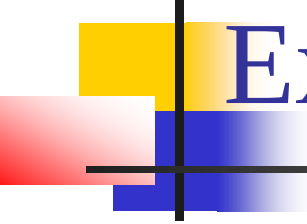
- Predict the label after learning w

$$\hat{y} = \text{sigmoid}(w^T x) = \text{sigmoid}(w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_d x_d)$$

- In classification tasks:

$$\hat{y} = \begin{cases} 1, & \sigma(w^T x) \geq 0.5 \\ 0, & \text{otherwise} \end{cases}$$

Where: $\sigma(w^T x) = \frac{1}{1 + e^{-w^T x}}$



Exercises

- Build a logistic regression model to predict exam results using the following training set

Hours	Pass	Hours	Pass
.5	0	2.75	1
.75	0	3	0
1	0	3.25	1
1.25	0	3.5	0
1.5	0	4	1
1.75	0	4.25	1
1.75	1	4.5	1
2	0	4.75	1
2.25	1	5	1
2.5	0	5.5	1